Rotations and angular momentum

Infinitesimal rotation about the z axis

To first order in the angle ϵ,

$$U_z(\epsilon) = I - \frac{\mathrm{i}\epsilon L_z}{\hbar} + O(\epsilon^2), \qquad U_z(-\epsilon) = U_z^\dagger(\epsilon), \qquad U_z^\dagger U_z = U_z U_z^\dagger = I.$$

At fixed (r, θ), a rotation acts on the azimuthal coordinate by $U_z(\epsilon)|\phi\rangle = |\phi + \epsilon\rangle$. *Exercise from the notes:* show that

$$\langle\phi|L_z|\psi\rangle = -\mathrm{i}\hbar\frac{\partial}{\partial\phi}\langle\phi \mid \psi\rangle.$$

Orbital angular momentum

$$\mathbf{L} = \mathbf{R} \times \mathbf{P} = L_x\hat{\mathbf{e}}_x + L_y\hat{\mathbf{e}}_y + L_z\hat{\mathbf{e}}_z, \qquad \begin{cases} L_x = YP_z - ZP_y, \\ L_y = ZP_x - XP_z, \\ L_z = XP_y - YP_x. \end{cases}$$

Using $[X_i, P_j] = \mathrm{i}\hbar\delta_{ij}$ and commuting distinct position or momentum components,

$$\begin{aligned} [L_x, L_y] &= [YP_z - ZP_y, ZP_x - XP_z] \\ &= [YP_z, ZP_x] + [ZP_y, XP_z] - [ZP_y, ZP_x] - [YP_z, XP_z]. \end{aligned}$$

The last two commutators vanish. For the others,

$$\begin{aligned} [YP_z, ZP_x] &= [YP_z, Z]\,P_x + Z\,[YP_z, P_x] = Y\,[P_z, Z]\,P_x = -\mathrm{i}\hbar YP_x, \\ [ZP_y, XP_z] &= [ZP_y, X]\,P_z + X\,[ZP_y, P_z] = X\,[Z, P_z]\,P_y = \mathrm{i}\hbar XP_y. \end{aligned}$$

Consequently,

$$[L_x, L_y] = \mathrm{i}\hbar L_z.$$

Cyclic permutation of x, y, z gives the compact form

$$\boxed{[L_i, L_j] = \mathrm{i}\hbar\sum_{k=1}^{3}\epsilon_{ijk}L_k.}$$

Operator cross products and rotational invariance

For an ordinary numerical vector, $\mathbf{a} \times \mathbf{a} = 0$. An operator vector need not have this property:

$$\mathbf{A} \times \mathbf{A} = [A_y, A_z]\,\hat{\mathbf{e}}_x + [A_z, A_x]\,\hat{\mathbf{e}}_y + [A_x, A_y]\,\hat{\mathbf{e}}_z.$$

Thus $\mathbf{A} \times \mathbf{A} = 0$ when all component commutators vanish. For example,

$$\mathbf{R} \times \mathbf{R} = 0, \qquad \mathbf{P} \times \mathbf{P} = 0, \qquad \mathbf{L} \times \mathbf{L} = \mathrm{i}\hbar\mathbf{L}.$$

The last equality follows component by component from $[L_y, L_z] = \mathrm{i}\hbar L_x$ and its cyclic partners.

Infinitesimal rotation about an arbitrary axis

For a unit vector $\hat{\mathbf{n}}$, put $\epsilon_i = \epsilon n_i$. Then

$$\begin{aligned} U_{\hat{\mathbf{n}}}(\epsilon) &= I - \frac{\mathrm{i}\epsilon}{\hbar}\mathbf{L}\cdot\hat{\mathbf{n}} + O(\epsilon^2) \\ &= I - \frac{\mathrm{i}}{\hbar}(L_x\epsilon_x + L_y\epsilon_y + L_z\epsilon_z) + O(\epsilon^2) \\ &= \left(I - \frac{\mathrm{i}L_x\epsilon_x}{\hbar}\right)\left(I - \frac{\mathrm{i}L_y\epsilon_y}{\hbar}\right)\left(I - \frac{\mathrm{i}L_z\epsilon_z}{\hbar}\right) + O(\epsilon^2) \\ &= U_x(\epsilon_x)U_y(\epsilon_y)U_z(\epsilon_z) + O(\epsilon^2). \end{aligned}$$

The equality between the different-axis product and the single-axis rotation is a first-order statement, since finite rotations do not in general commute.

Invariance of the Hamiltonian

A system is rotationally invariant if

$$U_{\hat{\mathbf{n}}}^{\dagger}(\epsilon)\,H\,U_{\hat{\mathbf{n}}}(\epsilon) = H \quad \text{for every axis and angle.}$$

This implies invariance about the three coordinate axes. Conversely, if the Hamiltonian is invariant under each coordinate-axis rotation, then to first order

$$\begin{aligned} U_{\hat{\mathbf{n}}}^{\dagger}HU_{\hat{\mathbf{n}}} &= U_z^{\dagger}U_y^{\dagger}U_x^{\dagger}HU_xU_yU_z + O(\epsilon^2) \\ &= U_z^{\dagger}U_y^{\dagger}HU_yU_z + O(\epsilon^2) \\ &= U_z^{\dagger}HU_z + O(\epsilon^2) = H + O(\epsilon^2). \end{aligned}$$

The infinitesimal condition generates the corresponding finite rotations.

Rotational invariance and compatible observables

Rotational invariance can be expressed through commutators:

$$\begin{aligned} U_{\hat{\mathbf{n}}}^{\dagger} H U_{\hat{\mathbf{n}}} = H \text{ for all } \hat{\mathbf{n}} &\iff [H, U_{\hat{\mathbf{n}}}] = 0 \text{ for all } \hat{\mathbf{n}} \\ &\iff [H, L_i] = 0 \quad (i = x, y, z). \end{aligned}$$

For an infinitesimal coordinate-axis rotation this follows from

$$0 = [H, I - \mathrm{i}\epsilon_i L_i/\hbar] = -\frac{\mathrm{i}\epsilon_i}{\hbar} [H, L_i] .$$

Writing $L^2 = L_x^2 + L_y^2 + L_z^2$, it also follows that $\left[H, L^2\right] = 0$.

Example: the free particle in three dimensions

$$H = \frac{\mathbf{P}^2}{2m}, \qquad E = \frac{p_x^2 + p_y^2 + p_z^2}{2m}.$$

Many momentum states have the same energy. Since

$$[H, P_x] = [H, P_y] = [P_x, P_y] = 0,$$

one may choose simultaneous eigenstates of H, P_x, P_y.

Clarification. These three labels alone do not always remove degeneracy: at fixed E, p_x, p_y, both signs of $p_z = \pm\sqrt{2mE - p_x^2 - p_y^2}$ are possible when the square root is nonzero. The full momentum vector provides an unambiguous alternative.

An angular-momentum choice

Different components of $\mathbf{L}$ are incompatible: $[L_i, L_j] \neq 0$ for distinct components in general. However, L^2 is rotationally invariant, and

$$\left[L^2, L_i\right] = 0.$$

Choosing the z axis conventionally,

$$\left[H, L^2\right] = [H, L_z] = \left[L^2, L_z\right] = 0.$$

Thus a simultaneous basis may be labeled $|E, \alpha, \beta\rangle$, where α and β are the eigenvalues of L^2 and L_z. Additional degeneracy labels are understood when required.

The same compatible set applies to a rotationally invariant central Hamiltonian

$$H = \frac{\mathbf{P}^2}{2m} + V(R), \qquad R = |\mathbf{R}| .$$

Angular-momentum eigenstates and ladder operators

Within a fixed energy subspace, suppress the energy label and write

$$L^2|\alpha,\beta\rangle = \alpha|\alpha,\beta\rangle, \qquad L_z|\alpha,\beta\rangle = \beta|\alpha,\beta\rangle.$$

Define $L_\pm = L_x \pm \mathrm{i}L_y$, so $L_+^\dagger = L_-$. Because $L^2 = L_x^2 + L_y^2 + L_z^2$,

$$\begin{aligned} L_-L_+ &= (L_x - \mathrm{i}L_y)(L_x + \mathrm{i}L_y) = L_x^2 + L_y^2 + \mathrm{i}\,[L_x, L_y] \\ &= L^2 - L_z^2 - \hbar L_z, \\ L_+L_- &= L^2 - L_z^2 + \hbar L_z. \end{aligned}$$

In particular,

$$L_-L_+|\alpha,\beta\rangle = (\alpha - \beta^2 - \hbar\beta)|\alpha,\beta\rangle.$$

Clarification. Here α is the eigenvalue of L^2, not its square root. Thus the intermediate α^2 in the handwriting is replaced by α.

The commutators are

$$\Big[L^2, L_\pm\Big] = 0, \qquad [L_z, L_\pm] = [L_z, L_x] \pm \mathrm{i}\,[L_z, L_y] = \pm\hbar L_\pm.$$

As in the harmonic-oscillator ladder construction,

$$\begin{aligned} L_zL_+|\alpha,\beta\rangle &= (L_+L_z + [L_z, L_+])|\alpha,\beta\rangle \\ &= (\beta + \hbar)L_+|\alpha,\beta\rangle, \\ L^2L_+|\alpha,\beta\rangle &= L_+L^2|\alpha,\beta\rangle = \alpha L_+|\alpha,\beta\rangle. \end{aligned}$$

Choosing a single irreducible ladder, with any additional labels suppressed, therefore gives

$$L_+|\alpha,\beta\rangle = C_+^{\alpha\beta}|\alpha,\beta+\hbar\rangle, \qquad L_-|\alpha,\beta\rangle = C_-^{\alpha\beta}|\alpha,\beta-\hbar\rangle.$$

The zero vector is allowed at either endpoint of the ladder.

The endpoints of an angular-momentum ladder

For a fixed α, successive applications of the ladder operators change β in steps of $\hbar$:

$$\cdots,\ \beta-2\hbar,\ \beta-\hbar,\ \beta,\ \beta+\hbar,\ \beta+2\hbar,\ \cdots.$$

Classically, $L_z^2 \le L^2$. Quantum mechanically, for a normalized state,

$$\begin{aligned}\alpha-\beta^2 &= \langle\alpha,\beta|L^2-L_z^2|\alpha,\beta\rangle \\ &= \langle\alpha,\beta|L_x^\dagger L_x|\alpha,\beta\rangle+\langle\alpha,\beta|L_y^\dagger L_y|\alpha,\beta\rangle \ge 0.\end{aligned}$$

Hence $\alpha \ge 0$ and $-\sqrt{\alpha} \le \beta \le \sqrt{\alpha}$. The ladder must terminate above and below:

$$L_+|\alpha,\beta_{\max}\rangle = 0, \qquad L_-|\alpha,\beta_{\min}\rangle = 0.$$

At the upper endpoint,

$$\begin{aligned}0 &= \langle\alpha,\beta_{\max}|L_-L_+|\alpha,\beta_{\max}\rangle \\ &= \alpha-\beta_{\max}^2-\hbar\beta_{\max},\end{aligned}$$

so $\alpha = \beta_{\max}(\beta_{\max}+\hbar)$. Similarly, using $L_+L_- = L^2-L_z^2+\hbar L_z$,

$$\alpha = \beta_{\min}(\beta_{\min}-\hbar).$$

Equating these expressions and using $\beta_{\max} \ge \beta_{\min}$ yields

$$\beta_{\min} = -\beta_{\max}, \qquad \beta_{\max} \ge 0.$$

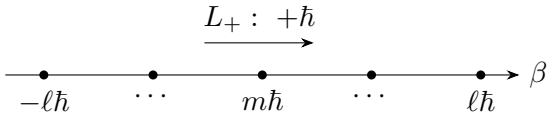
There is a nonnegative integer k such that

$$2\beta_{\max} = \beta_{\max}-\beta_{\min} = k\hbar.$$

Writing $\beta_{\max} = \ell\hbar$ gives

$$\ell = 0, \tfrac{1}{2}, 1, \tfrac{3}{2}, 2, \ldots,$$

the nonnegative integer and half-integer possibilities from the commutation algebra alone.



Integer and half-integer angular momentum

The endpoint result gives

$$\alpha = \ell\hbar(\ell\hbar + \hbar) = \ell(\ell+1)\hbar^2, \qquad \beta = m\hbar,$$

where $m = -\ell, -\ell+1, \ldots, \ell$ in steps of one. Relabeling the states,

$$L^2|\ell, m\rangle = \ell(\ell+1)\hbar^2|\ell, m\rangle, \qquad L_z|\ell, m\rangle = m\hbar|\ell, m\rangle.$$

The algebra allows $\ell = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$.

The orbital coordinate representation

Starting instead from $L_z = -\mathrm{i}\hbar\partial_\phi$,

$$-\mathrm{i}\hbar\frac{\partial}{\partial\phi}\langle\phi \mid \alpha, \beta\rangle = \beta\langle\phi \mid \alpha, \beta\rangle, \qquad \langle\phi \mid \alpha, \beta\rangle \propto \mathrm{e}^{\mathrm{i}\beta\phi/\hbar}.$$

For a scalar orbital wavefunction, single-valuedness under $\phi \mapsto \phi + 2\pi$ requires

$$\mathrm{e}^{2\pi\mathrm{i}\beta/\hbar} = 1 \quad \Longrightarrow \quad \beta = m\hbar, \qquad m \in \mathbb{Z}.$$

Thus the orbital representation restricts ℓ and m to integers, whereas the abstract commutation algebra also admits half-integers.

Why the algebra is more general

The total angular momentum is

$$\mathbf{J} = \mathbf{L} \otimes I_s + I_r \otimes \mathbf{S},$$

usually abbreviated as $\mathbf{J} = \mathbf{L} + \mathbf{S}$. Spin satisfies $\mathbf{S} \times \mathbf{S} = \mathrm{i}\hbar\mathbf{S}$; because spin and orbital operators act on different factors,

$$\mathbf{J} \times \mathbf{J} = \mathrm{i}\hbar\mathbf{J}.$$

Consequently, deriving eigenvalues only from this algebra does not specify that the operator must be $\mathbf{R} \times \mathbf{P}$. It includes spin and total angular momentum, for which half-integer representations occur.

Total angular momentum and ladder coefficients

For general angular momentum, use $\mathbf{J}$ instead of the orbital symbol $\mathbf{L}$:

$$J^2|j,m\rangle = j(j+1)\hbar^2|j,m\rangle, \qquad J_z|j,m\rangle = m\hbar|j,m\rangle.$$

For a spinless particle, $\mathbf{J} = \mathbf{L}$ and $j = \ell$, recovering the orbital formulas.

Suppose

$$J_\pm|j,m\rangle = C_\pm^{jm}|j,m\pm1\rangle, \qquad J_- = J_+^\dagger.$$

Taking the adjoint of the raising equation gives

$$\langle j,m|J_- = (C_+^{jm})^*\langle j,m+1|.$$

For normalized eigenstates,

$$\begin{aligned}\left|C_+^{jm}\right|^2 &= \langle j,m|J_-J_+|j,m\rangle \\ &= \langle j,m|J^2 - J_z^2 - \hbar J_z|j,m\rangle \\ &= \hbar^2\left[j(j+1) - m^2 - m\right] \\ &= \hbar^2(j-m)(j+m+1).\end{aligned}$$

Choose the relative phases so that these coefficients are real and nonnegative. The analogous calculation for lowering yields

$$C_+^{jm} = \hbar\sqrt{(j-m)(j+m+1)}, \qquad C_-^{jm} = \hbar\sqrt{(j+m)(j-m+1)}.$$

Therefore,

$$\boxed{J_\pm|j,m\rangle = \hbar\sqrt{(j\mp m)(j\pm m+1)}|j,m\pm1\rangle.}$$

Exercise retained from the notes

Calculate $\Delta L_x(t)$ and $\Delta L_y(t)$.

Clarification. A time-dependent answer requires a state and Hamiltonian. For a state $|\ell,m\rangle$ evolving only by an overall energy phase, $\langle L_x\rangle = \langle L_y\rangle = 0$ and $\Delta L_x = \Delta L_y = \hbar\sqrt{[\ell(\ell+1) - m^2]/2}$. This states the assumptions rather than assigning an unspecified time evolution.

Orbital eigenfunctions in position space

The highest-weight state satisfies $L_+|\ell,\ell\rangle = 0$. To use this in position space, first express the operators in spherical coordinates. With $\psi(\mathbf{r}) = \langle \mathbf{r} \mid \psi\rangle$,

$$\langle \mathbf{r}|L_x|\psi\rangle = -\mathrm{i}\hbar\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\psi(\mathbf{r}).$$

Clarification. The Cartesian prefactor is $-\mathrm{i}\hbar$, from $\mathbf{P} = -\mathrm{i}\hbar\boldsymbol{\nabla}$; the handwritten line has the opposite sign.

Using $y = r\sin\theta\sin\phi$, $z = r\cos\theta$,

$$\cos\phi = \frac{x}{\sqrt{x^2+y^2}}, \qquad \cos\theta = \frac{z}{r},$$

the chain rule gives

$$\begin{aligned} y\partial_z - z\partial_y &= r\sin\theta\sin\phi\left(\cos\theta\partial_r - \frac{\sin\theta}{r}\partial_\theta\right) \\ &\quad - r\cos\theta\left(\sin\theta\sin\phi\partial_r + \frac{\cos\theta\sin\phi}{r}\partial_\theta + \frac{\cos\phi}{r\sin\theta}\partial_\phi\right) \\ &= -\sin\phi\partial_\theta - \cos\phi\cot\theta\partial_\phi. \end{aligned}$$

Thus

$$\begin{aligned} L_x &= \mathrm{i}\hbar\left(\sin\phi\partial_\theta + \cos\phi\cot\theta\partial_\phi\right), \\ L_y &= \mathrm{i}\hbar\left(-\cos\phi\partial_\theta + \sin\phi\cot\theta\partial_\phi\right), \\ L_\pm &= \pm\hbar\mathrm{e}^{\pm\mathrm{i}\phi}\left(\partial_\theta \pm \mathrm{i}\cot\theta\partial_\phi\right). \end{aligned}$$

The L_y expression also follows by a $\pi/2$ azimuthal rotation of the L_x expression.
Since $L_z|\ell,\ell\rangle = \ell\hbar|\ell,\ell\rangle$, write $\langle r,\theta,\phi \mid \ell,\ell\rangle = u^\ell_\ell(r,\theta)\mathrm{e}^{\mathrm{i}\ell\phi}$. Then

$$(\partial_\theta - \ell\cot\theta)u^\ell_\ell = 0, \qquad \frac{\mathrm{d}u^\ell_\ell}{u^\ell_\ell} = \ell\frac{\cos\theta}{\sin\theta}\,\mathrm{d}\theta,$$

so

$$\psi^\ell_\ell(r,\theta,\phi) = R_\ell(r)\sin^\ell\theta\,\mathrm{e}^{\mathrm{i}\ell\phi}.$$

Exercise: starting from $L_-|\ell,-\ell\rangle = 0$, obtain $\psi^{-\ell}_\ell = R_\ell(r)\sin^\ell\theta\mathrm{e}^{-\mathrm{i}\ell\phi}$, up to a phase convention.

Normalized spherical harmonics

Define the highest-weight angular function by $Y_\ell^\ell(\theta,\phi) \propto \sin^\ell\theta e^{i\ell\phi}$. With

$$d^3r = r^2\,dr\;d\Omega, \qquad d\Omega = \sin\theta\;d\theta\;d\phi,$$

choose $\int \left|Y_\ell^\ell\right|^2 d\Omega = 1$. The needed polar integral is

$$\int_0^\pi \sin^{2\ell+1}\theta\;d\theta = \frac{2^{2\ell+1}(\ell!)^2}{(2\ell+1)!}.$$

Using the conventional Condon–Shortley phase gives

$$\boxed{Y_\ell^\ell(\theta,\phi) = \frac{(-1)^\ell}{2^\ell \ell!}\sqrt{\frac{(2\ell+1)!}{4\pi}}\sin^\ell\theta\, e^{i\ell\phi}.}$$

Generating the other magnetic states

For $\ell > 0$, the first lowering step gives

$$L_-|\ell,\ell\rangle = \hbar\sqrt{2\ell}|\ell,\ell-1\rangle.$$

Hence, using the differential operator from note 49,

$$Y_\ell^{\ell-1} = -\frac{e^{-i\phi}}{\sqrt{2\ell}}\left(\partial_\theta - i\cot\theta\partial_\phi\right)Y_\ell^\ell.$$

Before including its common normalization and phase, the differential operation on the highest-weight angular factor is

$$\begin{aligned}
&\left(\partial_\theta - i\cot\theta\partial_\phi\right)\left(\sin^\ell\theta e^{i\ell\phi}\right)\\
&\quad= \left(\ell\cos\theta\sin^{\ell-1}\theta + \ell\cos\theta\sin^{\ell-1}\theta\right)e^{i\ell\phi}\\
&\quad= 2\ell\cos\theta\sin^{\ell-1}\theta e^{i\ell\phi}.
\end{aligned}$$

Repeated lowering generates all Y_ℓ^m, called the spherical harmonics. One may also start at $m = -\ell$ and raise toward $m = 0$.

$$|\ell,\ell\rangle \xrightarrow{L_-} \cdots \xrightarrow{L_-} |\ell,0\rangle \xleftarrow{L_+} \cdots \xleftarrow{L_+} |\ell,-\ell\rangle.$$

The corresponding phase relation is

$$\boxed{Y_\ell^{-m}(\theta,\phi) = (-1)^m\left(Y_\ell^m(\theta,\phi)\right)^*.}$$

Clarification. The minus sign in the differential form of L_- is kept explicitly; it is required for consistency with the chosen Condon–Shortley phase and the positive ladder coefficients.

Central potentials and radial separation

A spherically symmetric Hamiltonian has the form

$$H = \frac{\mathbf{P}^2}{2m} + V(R), \qquad \mathbf{P}^2 \longleftrightarrow -\hbar^2\nabla^2.$$

The kinetic operator separates into radial and angular parts,

$$\mathbf{P}^2 = P_r^2 + \frac{L^2}{R^2}, \qquad P_r^2 \longleftrightarrow -\hbar^2 \frac{1}{r^2}\frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r}.$$

Here P_r^2 denotes the radial kinetic differential expression. For a position-space wavefunction,

$$\begin{aligned}\langle r,\theta,\phi|\mathbf{P}^2|\psi\rangle = &-\hbar^2\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right)\\ &-\frac{\hbar^2}{r^2}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]\psi.\end{aligned}$$

Thus the angular operator is

$$L^2 \longleftrightarrow -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right].$$

The three operators H, L^2, L_z commute, so choose states $|E,\ell,m\rangle$ with

$$\langle r,\theta,\phi|L^2|E,\ell,m\rangle = \ell(\ell+1)\hbar^2\langle r,\theta,\phi \mid E,\ell,m\rangle.$$

Writing $\Psi_{E\ell m}(r,\theta,\phi) = \langle r,\theta,\phi \mid E,\ell,m\rangle$, the Schrödinger equation becomes

$$\left[-\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)+\frac{\ell(\ell+1)\hbar^2}{2mr^2}+V(r)\right]\Psi_{E\ell m} = E\Psi_{E\ell m}.$$

The radial derivative acts only on r; the angular part is an L^2 eigenfunction. This motivates separating the radial function from the spherical harmonic.

The reduced radial wavefunction

Separate the simultaneous eigenfunction as

$$\Psi_{E\ell m}(r,\theta,\phi) = R_{E\ell}(r) Y_\ell^m(\theta,\phi).$$

The radial function obeys

$$\left\{ -\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}}{\mathrm{d}r}\right) - \frac{\ell(\ell+1)}{r^2}\right] + V(r) \right\} R_{E\ell}(r) = E R_{E\ell}(r).$$

The equation contains no magnetic quantum number m, so the radial function may be chosen independent of that label.

Normalization

Because the spherical harmonics are normalized,

$$\begin{aligned} 1 &= \int |\Psi_{E\ell m}(\mathbf{r})|^2 \,\mathrm{d}^3 r \\ &= \int_0^\infty r^2 \,|R_{E\ell}(r)|^2 \,\mathrm{d}r \underbrace{\int |Y_\ell^m(\theta,\phi)|^2 \,\mathrm{d}\Omega}_{1} \\ &= \int_0^\infty |r R_{E\ell}(r)|^2 \,\mathrm{d}r. \end{aligned}$$

Define the reduced radial function

$$U_{E\ell}(r) = r R_{E\ell}(r), \qquad R_{E\ell}(r) = \frac{U_{E\ell}(r)}{r}.$$

Then $\int_0^\infty |U_{E\ell}(r)|^2 \,\mathrm{d}r = 1$, and

$$\boxed{\left\{ \frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{2m}{\hbar^2}\left[E - V(r) - \frac{\ell(\ell+1)\hbar^2}{2mr^2}\right]\right\} U_{E\ell}(r) = 0.}$$

The centrifugal term acts as part of the effective one-dimensional radial potential. Absolute squares above also cover complex radial functions; ordinary squares are sufficient only when they are chosen real.

The hydrogen atom: radial quantization

Use the Coulomb convention of the notes, $V(r) = -e^2/r$. In SI units, each Coulomb factor e^2 here means $e_{\mathrm{SI}}^2/(4\pi\epsilon_0)$. With an infinitely heavy nucleus,

$$U'' + \frac{2m}{\hbar^2}\left[E + \frac{e^2}{r} - \frac{\ell(\ell+1)\hbar^2}{2mr^2}\right]U = 0.$$

For a bound state $E < 0$, set $\lambda = \sqrt{-2mE}/\hbar$. The limiting equations give

$$\begin{aligned} r \to \infty: &\quad U'' - \lambda^2 U \simeq 0, && U \propto \mathrm{e}^{-\lambda r}, \\ r \to 0: &\quad U'' - \frac{\ell(\ell+1)}{r^2}U \simeq 0, && U \propto r^{\ell+1} \quad \text{(regular solution)}. \end{aligned}$$

For $\ell = 0$, the regular boundary condition $U(0) = 0$ chooses the same behavior. Introduce

$$\rho = \lambda r, \qquad U(\rho) = \mathrm{e}^{-\rho} v(\rho), \qquad q = \frac{2me^2}{\hbar^2\lambda}.$$

The differential equation is

$$v'' - 2v' + \left[\frac{q}{\rho} - \frac{\ell(\ell+1)}{\rho^2}\right]v = 0.$$

Expand $v = \rho^{\ell+1}\sum_{k=0}^{\infty} c_k \rho^k$. Substitution gives

$$c_{k+1} = \frac{2(k+\ell+1) - q}{(k+1)(k+2\ell+2)}c_k.$$

Normalizable bound-state solutions require termination at some $K = 0, 1, 2, \ldots$, so $q = 2(K + \ell + 1)$. Therefore

$$E = -\frac{me^4}{2\hbar^2(K+\ell+1)^2}.$$

Clarification. The rescaling coefficient has been written explicitly as $q = 2me^2/(\hbar^2\lambda)$ to keep the dimensionless equation consistent with λ as an inverse length.

Define the principal quantum number $n = K + \ell + 1$. Then

$$\boxed{E_n = -\frac{me^4}{2\hbar^2 n^2}}, \qquad n = 1, 2, \ldots, \quad \ell = 0, 1, \ldots, n-1.$$

Each ℓ has $2\ell + 1$ magnetic states. The orbital degeneracy is $\sum_{\ell=0}^{n-1}(2\ell+1) = n^2$, before including spin.

The Rydberg energy and the Bohr-radius correspondence

The natural Coulomb energy scale is the Rydberg,

$$R_y = \frac{me^4}{2\hbar^2} \simeq 13.6057 \text{ eV}, \qquad E_n = -\frac{R_y}{n^2}.$$

The handwritten notes quote the numerical value 13.605 692 53(30) eV and attribute it to NIST; this is retained as a historical quotation, not as a current precision value.

Radial asymptotics

The source points to textbook pages 356–357 for explicit eigenfunctions, without identifying the textbook. Their large-radius behavior is

$$R_{n\ell}(r) \propto r^{n-1}\mathrm{e}^{-r/(na_0)} \qquad (r \to \infty),$$

whose power and exponential are independent of ℓ at fixed n. Indeed,

$$\lambda = \frac{\sqrt{-2mE_n}}{\hbar} = \frac{1}{n(\hbar^2/me^2)} = \frac{1}{na_0}, \qquad a_0 = \frac{\hbar^2}{me^2}.$$

Connection with the Bohr model

For the largest orbital angular momentum at a given n, take $\ell = n-1$. In that case the radial polynomial has no additional nodes, and

$$\Psi_{n,n-1,m}(\mathbf{r}) \propto r^{n-1}\mathrm{e}^{-r/(na_0)}Y^m_{n-1}(\theta,\phi).$$

The probability of finding the electron in a spherical shell of radius r and thickness $\mathrm{d}r$ is

$$\left(\int |\Psi(\mathbf{r})|^2\, r^2\, \mathrm{d}\Omega\right)\mathrm{d}r \propto r^{2n}\mathrm{e}^{-2r/(na_0)}\,\mathrm{d}r.$$

Thus the radial probability density has its maximum at

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r^{2n}\mathrm{e}^{-2r/(na_0)}\right) = 0 \quad \Longrightarrow \quad \boxed{r_{\max} = n^2 a_0.}$$

This equals the nth Bohr-orbit radius. The statement concerns the maximum of a radial probability distribution, rather than a definite electron trajectory.

Ehrenfest's theorem and the virial theorem

For an operator Ω with no explicit time dependence, the Heisenberg equation implies

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}\langle\Omega\rangle = \langle[\Omega, H]\rangle.$$

In an energy eigenstate,

$$\begin{aligned}\langle E|\,[\Omega, H]\,|E\rangle &= \langle E|\Omega H|E\rangle - \langle E|H\Omega|E\rangle \\ &= E\big(\langle E|\Omega|E\rangle - \langle E|\Omega|E\rangle\big) = 0.\end{aligned}$$

Consequently all well-defined expectation values of such operators are stationary in an energy eigenstate.

Choose $\Omega = \mathbf{R}\cdot\mathbf{P}$

For $H = \mathbf{P}^2/(2m) + V(\mathbf{R})$, the operator equations give

$$\begin{aligned}\dot{X}_i &= \frac{1}{\mathrm{i}\hbar}\,[X_i, H] = \frac{1}{2m\mathrm{i}\hbar}\left[X_i, P_i^2\right] = \frac{P_i}{m}, \\ \dot{P}_i &= \frac{1}{\mathrm{i}\hbar}\,[P_i, H] = \frac{1}{\mathrm{i}\hbar}\,[P_i, V] = -\frac{\partial V}{\partial X_i}.\end{aligned}$$

In vector notation, $\dot{\mathbf{R}} = \mathbf{P}/m$ and $\dot{\mathbf{P}} = -\boldsymbol{\nabla}_R V$. Therefore a stationary state satisfies

$$0 = \left\langle \dot{\mathbf{R}}\cdot\mathbf{P}\right\rangle + \left\langle \mathbf{R}\cdot\dot{\mathbf{P}}\right\rangle = \frac{\langle\mathbf{P}^2\rangle}{m} - \langle\mathbf{R}\cdot\boldsymbol{\nabla}_R V\rangle.$$

If $V(\mathbf{R}) = cR^k = c(X_1^2 + X_2^2 + X_3^2)^{k/2}$, then

$$-\frac{\partial V}{\partial X_i} = -ckX_i R^{k-2}, \qquad \mathbf{R}\cdot\boldsymbol{\nabla}_R V = ckR^k = kV.$$

Hence

$$\boxed{\langle T\rangle = \left\langle\frac{\mathbf{P}^2}{2m}\right\rangle = \frac{k}{2}\langle V\rangle.}$$

This is the virial theorem for a homogeneous power-law potential, for stationary states where the indicated expectations and boundary conditions are well defined.

The Coulomb virial relation and the mean inverse radius

For the Coulomb potential, take $k = -1$ and $c = -e^2$. The virial theorem becomes

$$\left\langle \frac{\mathbf{P}^2}{2m} \right\rangle = -\frac{1}{2} \left\langle -\frac{e^2}{R} \right\rangle .$$

For a normalized hydrogenic energy eigenstate $|n, \ell, m\rangle$,

$$\begin{aligned} E_n &= \left\langle \frac{\mathbf{P}^2}{2m} - \frac{e^2}{R} \right\rangle_{n\ell m} \\ &= \langle T \rangle_{n\ell m} + \langle V \rangle_{n\ell m} \\ &= \frac{1}{2} \langle V \rangle_{n\ell m} = -\frac{e^2}{2} \left\langle \frac{1}{R} \right\rangle_{n\ell m} . \end{aligned}$$

On the other hand,

$$E_n = -\frac{me^4}{2\hbar^2 n^2} = -\frac{e^2}{2a_0 n^2}.$$

Comparing the two expressions gives

$$\boxed{\left\langle \frac{1}{R} \right\rangle_{n\ell m} = \frac{1}{n^2 a_0}.}$$

This expectation is independent of ℓ and m in the nonrelativistic Coulomb problem considered here.

Clarification. The potential contribution to H is $+V = -e^2/R$. The intermediate handwritten expression that subtracts $\langle -e^2/R \rangle$ is a sign slip; the final result in the notes is consistent with the corrected addition above.

Atomic energy, length, and speed scales

With the same Coulomb-unit convention as the preceding pages,

$$R_y = \frac{me^4}{2\hbar^2} = \frac{1}{2}mc^2\left(\frac{e^2}{\hbar c}\right)^2 = \frac{1}{2}mc^2\alpha^2,$$
$$\alpha = \frac{e^2}{\hbar c}, \qquad \alpha^{-1} \simeq 137.036.$$

Clarification. The source's correction callout explicitly changes the printed label to $1/\alpha$. The number 137.036 is the approximate inverse fine-structure constant, not α itself.

The rest-energy estimates written in the notes are

$$mc^2 \simeq 0.511 \text{ MeV} \simeq 0.5 \text{ MeV} \quad \text{(electron)},$$
$$Mc^2 \simeq 938.3 \text{ MeV} \simeq 1000 \text{ MeV} \quad \text{(proton)}.$$

These rounded values illustrate the separation of atomic and rest-energy scales.

The Bohr length and the decay constant

$$R_y = \frac{e^2}{2a_0}, \qquad a_0 = \frac{\hbar^2}{me^2}, \qquad \frac{1}{na_0} = \sqrt{\frac{-2mE_n}{\hbar^2}}.$$

The Coulomb virial result gives

$$\langle T\rangle_{n\ell m} = \frac{R_y}{n^2}.$$

For the ground state in the nonrelativistic description,

$$\frac{1}{2}m\langle v^2\rangle_{n=1} = \frac{1}{2}mc^2\alpha^2, \qquad v_{\text{rms}} = \sqrt{\langle v^2\rangle_{n=1}} = c\alpha.$$

Using the conventional dimensionless speed $\beta = v/c$,

$$\beta_{\text{rms}} = \alpha, \qquad R_y = \frac{1}{2}mc^2\alpha^2 = \frac{1}{2}mc^2\beta_{\text{rms}}^2.$$

Spin and the product Hilbert space

For a spin-s system, the magnetic quantum number takes the values

$$m_s = -s, -s+1, \ldots, s-1, s,$$

giving $2s+1$ spin states, with S_z eigenvalues $m_s\hbar$. The spatial Hilbert space and spin Hilbert space are separate factors. For a product state, write

$$|\Psi\rangle = |\psi\rangle \otimes |\chi\rangle,$$

where $|\psi\rangle$ is spatial and $|\chi\rangle$ carries the spin.

A discrete spatial basis

For spatial basis vectors $|x_1\rangle, \ldots, |x_N\rangle$, the product basis coefficient is

$$(\langle x_a| \otimes \langle m_s|)(|\psi\rangle \otimes |\chi\rangle) = \langle x_a \mid \psi\rangle\langle m_s \mid \chi\rangle.$$

Ordering the coefficients first by spin label gives the column

$$\begin{bmatrix} \langle x_1 \mid \psi\rangle\langle s \mid \chi\rangle \\ \vdots \\ \langle x_N \mid \psi\rangle\langle s \mid \chi\rangle \\ \langle x_1 \mid \psi\rangle\langle s-1 \mid \chi\rangle \\ \vdots \\ \langle x_N \mid \psi\rangle\langle -s \mid \chi\rangle \end{bmatrix}.$$

A continuous spatial representation

On replacing the discrete position labels by a continuous x, the same state is represented by a $(2s+1)$-component wavefunction,

$$\boldsymbol{\Psi}(x) = \begin{bmatrix} \langle x \mid \psi\rangle\langle s \mid \chi\rangle \\ \langle x \mid \psi\rangle\langle s-1 \mid \chi\rangle \\ \vdots \\ \langle x \mid \psi\rangle\langle -s \mid \chi\rangle \end{bmatrix}, \qquad \Psi_{m_s}(x) = \psi(x)\chi_{m_s}.$$

The source also writes the product as $\psi(x)\chi_m$.

Clarification. The factorized expression describes a product state. A general spin-position state may require independent component functions $\Psi_{m_s}(x)$ and need not factor into one common $\psi(x)$ times a spin vector.

Spin commutators and the spin-one-half matrices

The full angular-momentum operator is

$$\mathbf{J} = \mathbf{L} \otimes I_s + I_r \otimes \mathbf{S}, \qquad [J_i, J_j] = \mathrm{i}\hbar \sum_k \epsilon_{ijk} J_k.$$

The abbreviated commutator $[L_i, S_j]$ means

$$[L_i \otimes I_s, I_r \otimes S_j] = 0,$$

because the two operators act on different factors. Expanding $[L_i + S_i, L_j + S_j]$, the mixed terms vanish. Subtracting the orbital commutator from the total one gives

$$\boxed{[S_i, S_j] = \mathrm{i}\hbar \sum_k \epsilon_{ijk} S_k}, \qquad \mathbf{S} \times \mathbf{S} = \mathrm{i}\hbar \mathbf{S}.$$

An electron has $s = \frac{1}{2}$

The two S_z eigenvalues are $+\hbar/2$ and $-\hbar/2$. In the ordered basis $|+\rangle = |\frac{1}{2}, \frac{1}{2}\rangle$, $|-\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle$,

$$S_z = \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The ladder formula supplies the individual matrix entries,

$$\langle \tfrac{1}{2}, m' | S_\pm | \tfrac{1}{2}, m \rangle = \hbar \sqrt{(\tfrac{1}{2} \mp m)(\tfrac{1}{2} \pm m + 1)}\, \delta_{m', m \pm 1}.$$

Consequently,

$$S_+ = \hbar \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad S_- = \hbar \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$$

or equivalently $S_\pm = (\hbar/2) \begin{bmatrix} 0 & 1 \pm 1 \\ 1 \mp 1 & 0 \end{bmatrix}$. It follows that

$$S_x = \tfrac{1}{2}(S_+ + S_-) = \frac{\hbar}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad S_y = \tfrac{1}{2\mathrm{i}}(S_+ - S_-) = \frac{\hbar}{2} \begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix}.$$

The dimensionless matrices $\sigma_i = 2S_i/\hbar$ are the Pauli matrices; the S_i themselves retain the units of angular momentum.

Position-spin wavefunctions and momentum

For a product state $|\Psi\rangle = |\varphi\rangle \otimes |\chi\rangle$, the S_z representation first replaces the spin factor by its two coefficients:

$$\begin{bmatrix} \langle + \mid \chi \rangle |\varphi\rangle \\ \langle - \mid \chi \rangle |\varphi\rangle \end{bmatrix}.$$

Taking spatial position matrix elements gives the position-plus-S_z representation,

$$\begin{aligned} \boldsymbol{\Psi}(\mathbf{r}) &= \begin{bmatrix} \langle + \mid \chi \rangle \langle \mathbf{r} \mid \varphi \rangle \\ \langle - \mid \chi \rangle \langle \mathbf{r} \mid \varphi \rangle \end{bmatrix} = \begin{bmatrix} \Psi_+(\mathbf{r}) \\ \Psi_-(\mathbf{r}) \end{bmatrix} \\ &= \Psi_+(\mathbf{r}) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \Psi_-(\mathbf{r}) \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \end{aligned}$$

The two-component expression also represents general, nonfactorized states when the two functions are allowed to vary independently.

Momentum acts on the spatial factor

In the product Hilbert space,

$$\mathbf{P}_{r+s} = \mathbf{P}_r \otimes I_s.$$

Therefore its action in the position-plus-S_z representation is

$$\mathbf{P}_{r+s} \;\longleftrightarrow\; \begin{bmatrix} -\mathrm{i}\hbar\boldsymbol{\nabla} & 0 \\ 0 & -\mathrm{i}\hbar\boldsymbol{\nabla} \end{bmatrix},$$

so

$$(\mathbf{P}\boldsymbol{\Psi})(\mathbf{r}) = \begin{bmatrix} -\mathrm{i}\hbar\boldsymbol{\nabla}\Psi_+(\mathbf{r}) \\ -\mathrm{i}\hbar\boldsymbol{\nabla}\Psi_-(\mathbf{r}) \end{bmatrix}.$$

The gradient does not mix spin-up and spin-down components.

Fixed spin and normalization of a spinor wavefunction

In the spin-up/down basis, the entries of S^2 are

$$\begin{bmatrix} \langle +|S^2|+\rangle & \langle +|S^2|-\rangle \\ \langle -|S^2|+\rangle & \langle -|S^2|-\rangle \end{bmatrix} = \hbar^2 \begin{bmatrix} \frac{1}{2}(\frac{1}{2}+1) & 0 \\ 0 & \frac{1}{2}(\frac{1}{2}+1) \end{bmatrix} = \frac{3}{4}\hbar^2 I_s.$$

Thus S^2 acts as the same constant on every state of the spin-one-half representation. More generally, intrinsic spin labels an elementary particle through the eigenvalue $s(s+1)\hbar^2$, in the same sense that charge and rest mass are intrinsic labels. In contrast, the orbital quantum number ℓ can vary between spatial states of the same particle.

Normalization in space and spin

Insert the complete product basis into $\langle \Psi \mid \Psi \rangle$:

$$\begin{aligned} 1 &= \langle \Psi \mid \Psi \rangle \\ &= \int \mathrm{d}^3 r \sum_{m_s=\pm 1/2} \langle \Psi \mid \mathbf{r}, m_s \rangle \langle \mathbf{r}, m_s \mid \Psi \rangle \\ &= \int \mathrm{d}^3 r \sum_{m_s=\pm 1/2} |\langle \mathbf{r}, m_s \mid \Psi \rangle|^2 \\ &= \int \left[|\Psi_+(\mathbf{r})|^2 + |\Psi_-(\mathbf{r})|^2 \right] \mathrm{d}^3 r. \end{aligned}$$

Equivalently,

$$1 = \int \mathbf{\Psi}^\dagger(\mathbf{r}) \mathbf{\Psi}(\mathbf{r}) \, \mathrm{d}^3 r = \int \begin{bmatrix} \Psi_+^*(\mathbf{r}) & \Psi_-^*(\mathbf{r}) \end{bmatrix} \begin{bmatrix} \Psi_+(\mathbf{r}) \\ \Psi_-(\mathbf{r}) \end{bmatrix} \mathrm{d}^3 r.$$

Clarification. The adjoint is a row and the ket representation is a column. Their order above gives a scalar norm; the reversed column-times-row sketch in the source would instead give an outer product matrix.

Spin and orbital operators on two-component wavefunctions

The three Pauli matrices are

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \sigma_y = \begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix}, \qquad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad S_i = \frac{\hbar}{2}\sigma_i.$$

Spin acts only on the two spin labels. In components,

$$\langle \mathbf{r}, m_s | S_i | \Psi \rangle = \sum_{m_s'} \langle m_s | S_i | m_s' \rangle \langle \mathbf{r}, m_s' \mid \Psi \rangle.$$

In column notation this is simply

$$(S_i \boldsymbol{\Psi})(\mathbf{r}) = \frac{\hbar}{2}\sigma_i \begin{bmatrix} \Psi_+(\mathbf{r}) \\ \Psi_-(\mathbf{r}) \end{bmatrix},$$

where

$$\Psi_+(\mathbf{r}) = \langle \mathbf{r}, +\tfrac{1}{2} \mid \Psi \rangle, \qquad \Psi_-(\mathbf{r}) = \langle \mathbf{r}, -\tfrac{1}{2} \mid \Psi \rangle.$$

Orbital operators are diagonal in spin space

For instance,

$$(L_z \boldsymbol{\Psi})(\mathbf{r}) = \begin{bmatrix} -\mathrm{i}\hbar\partial_\phi & 0 \\ 0 & -\mathrm{i}\hbar\partial_\phi \end{bmatrix} \begin{bmatrix} \Psi_+(\mathbf{r}) \\ \Psi_-(\mathbf{r}) \end{bmatrix}.$$

Likewise L^2, L_x, L_y, and $L_\pm$ each act by the corresponding orbital differential operator on both components, without mixing the spin labels:

$$(A_r \otimes I_s)\boldsymbol{\Psi} = \begin{bmatrix} A_r \Psi_+ \\ A_r \Psi_- \end{bmatrix} \quad \text{for } A_r \in \{L^2, L_x, L_y, L_z, L_+, L_-\}.$$

Independent orbital and spin degrees of freedom

Suppose the Hamiltonian separates into independent spatial and spin parts,

$$H = H_0 \otimes I_s + I_r \otimes H_s,$$

often abbreviated to $H = H_0 + H_s$. Here H_0 contains only spatial operators, and H_s contains only spin operators.

By contrast, an interaction H_{so} involving both sectors cannot generally be written as such a sum. A familiar example is

$$H = H_0 + H_s + H_{so}, \qquad H_{so} = \xi\, \mathbf{L} \cdot \mathbf{S},$$

with a coupling coefficient ξ (which can depend on radius in a central-field model).

Separated time evolution

For time-independent independent parts, a product initial state stays factorized:

$$|\Psi(t)\rangle = |\psi_0(t)\rangle \otimes |\chi(t)\rangle,$$

where

$$|\psi_0(t)\rangle = \mathrm{e}^{-\mathrm{i}H_0 t/\hbar}|\psi_0(0)\rangle, \qquad |\chi(t)\rangle = \mathrm{e}^{-\mathrm{i}H_s t/\hbar}|\chi(0)\rangle.$$

Clarification. The spin propagator acts on the initial spin ket $|\chi(0)\rangle$. A factorized Hamiltonian alone does not imply that every initial state is a product; an initially entangled state still evolves by the product of the two propagators.

The spin factor alone

For spin one half,

$$|\chi\rangle = \alpha|+\rangle + \beta|-\rangle \quad \longleftrightarrow \quad \begin{bmatrix}\alpha\\ \beta\end{bmatrix}, \qquad |+\rangle \longleftrightarrow \begin{bmatrix}1\\ 0\end{bmatrix}, \quad |-\rangle \longleftrightarrow \begin{bmatrix}0\\ 1\end{bmatrix},$$

with $\alpha = \langle + \mid \chi\rangle$ and $\beta = \langle - \mid \chi\rangle$. Inserting the spin completeness relation gives

$$\begin{aligned} 1 = \langle \chi \mid \chi\rangle &= \sum_{m_s=\pm 1/2} \langle \chi \mid m_s\rangle\langle m_s \mid \chi\rangle \\ &= |\langle + \mid \chi\rangle|^2 + |\langle - \mid \chi\rangle|^2 = |\alpha|^2 + |\beta|^2 . \end{aligned}$$